PEER REVIEW HISTORY

BMJ Open publishes all reviews undertaken for accepted manuscripts. Reviewers are asked to complete a checklist review form (**http://bmjopen.bmj.com/site/about/resources/checklist.pdf**) and are provided with free text boxes to elaborate on their assessment. These free text comments are reproduced below.

ARTICLE DETAILS

TITLE (PROVISIONAL)	Prevalence of malnutrition among old age people in Africa: Systematic Review and Meta-Analysis
AUTHORS	Seid, Ahmed; Fentahun, Netsanet

VERSION 1 – REVIEW

REVIEWER	Wondmieneh, Adam Woldia University, Nursing
REVIEW RETURNED	18-Jun-2022

GENERAL COMMENTS	Manuscript ID: bmjopen-2022-065197 Title: Prevalence of malnutrition among old age people in Africa: Systematic Review and Meta-Analysis Thank you for the opportunity to review the manuscript on the Prevalence of malnutrition among old age people in Africa: Systematic Review and Meta-Analysis The article is well-written and represents interesting findings. However, this article estimate the pooled prevalence of malnutrition in Africa, but it is better if the author shows undernutrition or over nutrition among old African population, because it is important for intervention and recommendation. Methods 1. Even though you couldn't clearly explain your outcome variable, in this systematic review and meta-analysis, the prevalence of malnutrition among old peoples was determined by dividing the number of patients having malnutrition to the total number of study participants included in the final analysis. You included one case control study in this meta-analysis. However, you can't calculate prevalence from a case control study. Because the way you select the sample based on the a certain number of cases according to your sample size calculation and you select the control based on matched with the cases or unmatched with a ratio 1:1 or until 1: 3. 2. Please indicate whether the protocol for the review was registered on PROSPERO database. If registered, what was the number? 3. Since this study is generalized to African people, what is the reason to exclude studies published by other African union working languages? Some important data might be obtained from other languages. For example North African's papers are published in Arabic language, but you exclude these articles. 4. Outcome measurement tool needs explanation? For example what is the cut of point to say undernutrition, over-weight and obese based on BMI, MUAC or other nutritional assessment tools? 5. Was the goodness of fit of the model assessed? How to assess? Result section

	1. Please show the estimated pooled prevalence of undernutrition and over nutrition among old people separately? 2. The author stated high level heterogeneity of was detected between the included studies, what do you did to detect source of heterogeneity? 3. In this study there is publication bias, and the source of publication bas may be due to small study, there for why not you conduct trim fill analysis? Conclusion This paper does not have recommendations; based on the findings of your study you should recommend the concerned body.

REVIEWER	Omar, Mohamed Jigjiga University
REVIEW RETURNED	26-Jul-2022

GENERAL COMMENTS	Comments in general 1. Your research is in a critical area of public health. It is clear that the authors' first language is not English. As a result, the language and grammar are substandard, and there are several spelling errors throughout the work, making it difficult to comprehend. Professional language editing and a spelling and grammar check would greatly benefit the article. While this paper is intriguing, it has a number of flaws. The sample size for the population study is somewhat small. Comments in specific Introduction 2. Starting with malnutrition as a global problem, then discussing its previous reports in the world across many countries, and probable consequences, and finally narrowing it down to Africa, will greatly help your introduction. Then you can describe the current evidence gap, which will lead to a rationale for your study which is not clear, followed by the study's objectives. 3. You attempted to do so in the introduction, but it has to be reorganized because the global data appears late in your introduction. Methods 4. A protocol and registration are required for systematic reviews and meta-analyses. 5. Justify your decision to solely include observational research. Most you used Cross-sectional studies which provides more persuasive data; it's unclear why they weren't included other study designs. 6. Systematic reviews exclude case studies, reviews, theses, and letters to editors, editorials, commentaries, and conference abstracts. These should be included as exclusion criteria by the writers. 7. These relationships have been described in the literature, but they are significant at a national level in Africa to increase malnutrition among elderly knowledge and inform future research and intervention programs. 8. Operational definition, better to integrate into the content of manuscript and explained as you proceed with writing Results 9. The document must include a PRISMA flow diagram reference to confirm. 10. How you did sensitivity analysis, please write the code you run here Discussions

	11. Discussion (in general, there is a lack of in-depth explanation and substantiation of associations in the discussion.) 12. Come up with a strong conclusion and especially recommendation from the study which can aid scientific continuity and draw attention to guideline developments and policy makers captivating. References 13. Some references are incomplete, some example… please correct them Keep consistence in using lower or upper case The referencing style is not consistent (advised to use a reference management software) Please revise it

VERSION 1 – AUTHOR RESPONSE

Raised comments	**Our responses**
Reviewer #1	
The article is well-written and represents interesting findings.	We thank the reviewer for his careful reading of the paper and appreciation!
However, this article estimates the pooled prevalence of malnutrition in Africa, but it is better if the author shows undernutrition or over nutrition among old African population, because it is important for intervention and recommendation.	Thank you for this comment. We agree with this comment. Therefore, we have discussed them separately in the manuscript (under page 7&8)
Methods 1. Even though you couldn't clearly explain your outcome variable, in this systematic review and meta-analysis, the prevalence of malnutrition among old peoples was determined by dividing the number of patients having malnutrition to the total number of study participants included in the final analysis. You included one case control study in this meta-analysis. However, you can't calculate prevalence from a case control study. Because the way you select the sample based on the a certain number of cases according to your sample size calculation and you select the control based on matched with the cases or unmatched with a ratio 1:1 or until 1: 3.	We thank the reviewer for this close attention to the evidence on which our findings are based. However, we have stated and explained the outcome variable in the inclusion criteria as the prevalence and/or incidence of malnutrition among old age people regardless of the study design. Nonetheless, to respond to the reviewer's comments, we calculated the proportion of malnutrition from the reported study as the total results of undernutrition per the total sample size of the study without classification of cases and controls. Our primary goal is to determine what proportion of the study population was undernourished, normal, or over nourished by combining the cases and controls.
2. Please indicate whether the protocol for the review was registered on PROSPERO database. If registered, what was the number?	We thank the reviewer for this point. Unfortunately, we did not register this review under PROSPERO. Yet, prior to the beginning of this systematic review, the databases of Cochrane Library, PROSPERO, and JBI were searched and no previous systematic review on this specific topic was identified. We also believe that it is not mandatory.

3. Since this study is generalized to African people, what is the reason to exclude studies published by other African union working languages? Some important data might be obtained from other languages. For example North African's papers are published in Arabic language, but you exclude these articles.	Thank you for this comment. We did not exclude published studies for African elderly people. However, language restrictions were used to search for studies, and only English was used. We have also discussed this under the study limitation.
4. Outcome measurement tool needs explanation? For example what is the cut of point to say undernutrition, over-weight and obese based on BMI, MUAC or other nutritional assessment tools?	We appreciate the reviewer's focus on the data that supports our results. However, we have discussed the outcome measurement tools in a separate title under the method section. As we have explained there, we included studies that measured malnutrition with any type of tool. We used the reported prevalence of malnutrition based on the standard definition set for each nutritional assessment method. So, we do not worry about the cutoff points but simply take the reported outcome.
5. Was the goodness of fit of the model assessed? How to assess?	Thank you for this concern. Yes, we have assessed the goodness of fit via random effect model. We have also put a reference # (26) for further details.
Result section 1. Please show the estimated pooled prevalence of undernutrition and over nutrition among old people separately?	Thank you for this comment. We have done it as commented (page 7&8)
2. The author stated high level heterogeneity of was detected between the included studies, what do you did to detect source of heterogeneity?	Thank you for this question. Subgroup analysis and meta-regression were performed to detect source of heterogeneity. We also discussed these in the result section (page 9 & 10)
3. In this study there is publication bias, and the source of publication bas may be due to small study, there for why not you conduct trim fill analysis?	Thank you for this important question. The Trim and Fill analysis was performed. Now, we discussed these in the result section as per the comment.
Conclusion This paper does not have recommendations; based on the findings of your study you should recommend the concerned body.	We thank the reviewer for this point. If the reviewer carefully read the manuscript, we have a recommendation for the concerned body. We have stated that "due attention to geriatric nutrition is mandatory, and to better monitor and evaluate, the use of a more reliable, valid, and standardized malnutrition assessment tool is strongly recommended." This is found under the abstract conclusion part as well as the final conclusion part of the document.

Raised comments	**Our responses**
Reviewer #2	
Comments in general 1. Your research is in a critical area of public health.	We thank the reviewer for his appreciation!

It is clear that the authors' first language is not English. As a result, the language and grammar are substandard, and there are several spelling errors throughout the work, making it difficult to comprehend. Professional language editing and a spelling and grammar check would greatly benefit the article. While this paper is intriguing, it has a number of flaws.	We thank the reviewer for paying close attention to the paper. We tried to use professional language editing and grammar checking to correct our writing style before manuscript submission. However, if there is/are any error(s), please notify us so that we can correct it (them).
The sample size for the population study is somewhat small.	We appreciate the reviser's comments. However, sample size is not first calculated for this study since it is a review type. We tried to include all studies depending on the inclusion criteria. And finally, we get 37 and 36 articles for qualitative (systematic review) and quantitative analysis (meta-analysis), respectively. We think these are more than enough for this type of study.
Comments in specific **Introduction** 2. Starting with malnutrition as a global problem, then discussing its previous reports in the world across many countries, and probable consequences, and finally narrowing it down to Africa, will greatly help your introduction. Then you can describe the current evidence gap, which will lead to a rationale for your study which is not clear, followed by the study's objectives. 3. You attempted to do so in the introduction, but it has to be reorganized because the global data appears late in your introduction.	Thank you for this comment. We have attempted to rearrange and describe the world situation in the introduction part to make it more-clear. However, we have nonetheless decided to reduce the amount of information provided to respect the word limit.
Methods 4. A protocol and registration are required for systematic reviews and meta-analyses.	We thank the reviewer for this point. Unfortunately, we did not register this review under PROSPERO. Yet, prior to the beginning of this systematic review, the databases of Cochrane Library, PROSPERO, and JBI were searched and no previous systematic review on this specific topic was identified. We also believe that it is not mandatory.
5. Justify your decision to solely include observational research. Most you used Cross-sectional studies which provides more persuasive data; it's unclear why they weren't included other study designs.	We thank the reviewer for this close attention to the evidence on which our findings are based. However, we have stated and explained the inclusion criteria as "all studies that report the prevalence and/or incidence of malnutrition among old age people regardless of the study design." (Page 4) Nonetheless, if the nutritional status was measured after a nutrition intervention study, it was excluded as this was not considered to reflect the initial prevalence of malnutrition.

6. Systematic reviews exclude case studies, reviews, theses, and letters to editors, editorials, commentaries, and conference abstracts. These should be included as exclusion criteria by the writers. 7. These relationships have been described in the literature, but they are significant at a national level in Africa to increase malnutrition among elderly knowledge and inform future research and intervention programs.	Thank you for your comments. Yes, we have excluded all these and we stated them under the exclusion criteria (page 4).
8. Operational definition, better to integrate into the content of manuscript and explained as you proceed with writing.	We thank the reviewer for this comment. However, we did not get an operational definition in the BMJ open guidelines. If it is necessary, we will add it.
Results 9. The document must include a PRISMA flow diagram reference to confirm.	Thank you for your comments. Yes, we have it and it was attached as Figure legends.
10. How you did sensitivity analysis, please write the code you run here	Thank you for your comments. Yes, we did sensitivity analysis. Since we used STATA, here is the command: metaninf logp selogp, label(namevar= authors , yearvar=years) random texts(110) eform
Discussions 11. Discussion (in general, there is a lack of in-depth explanation and substantiation of associations in the discussion.) 12. Come up with a strong conclusion and especially recommendation from the study which can aid scientific continuity and draw attention to guideline developments and policy makers captivating.	We thank the reviewer for this close attention to paper. We have tried to make it as per the comments.
References 13. Some references are incomplete, some example… please correct them Keep consistence in using lower or upper case The referencing style is not consistent (advised to use a reference management software) Please revise it.	We thank the reviewer for this point. We have tried to make it as per the comments. We used Mendeley Reference Manager. However, if there is/are any error(s), please notify us so that we can correct it (them).

VERSION 2 – REVIEW

REVIEWER	Omar, Mohamed Jigjiga University
REVIEW RETURNED	22-Sep-2022

GENERAL COMMENTS	No comments are required at this stage.